

Project Design Phase Problem – Solution

Date	30 september 2026
Team ID	SWTID-2026-2683
Project Name	Script-Controlled ACL – Restrict Record Access Based on Field Value
Maximum Marks	2 Marks

Problem – Solution :

The Problem-Solution Fit means that a real problem has been found for the customer and that the proposed solution actually solves it. For this project, the problem is that users can see or change records they should not, and the solution is a set of script-controlled ACLs that restrict.

1. CUSTOMER SEGMENT(S) [CS] - Organisations (schools, colleges, enterprises) storing sensitive records on ServiceNow - ServiceNow administrators and developers - Data owners such as faculty or branch coordinators - Learners practising ACL security	6. CUSTOMER CONSTRAINTS [CC] - Limited ServiceNow security knowledge - Risk of locking out valid users or exposing data if ACLs are wrong - Need a simple solution with no extra plugins or cost - Limited time and small test data	5. AVAILABLE SOLUTIONS [AS] - Role-only ACLs: simple, but cannot filter by field value - Data-condition-only ACLs: filter records, limited logic - Separate tables or views per branch: duplicate data, hard to maintain - Manual sharing or spreadsheets: no real control
2. JOBS-TO-BE-DONE / PROBLEMS [J&P] - Let only EEE-branch users view EEE records; admins keep full access - Control create, edit and delete separately from viewing - Prove the restrictions work for each user type	9. PROBLEM ROOT CAUSE [RC] - Table access is broad by default unless ACLs are written - Roles alone do not know about record content - Read, create, write and delete each need their own rule	7. BEHAVIOUR [BE] - Users open the list or form and expect to see or edit their records - Admins create roles and ACLs, then impersonate users to test - Users without access report missing records or a missing New button
3. TRIGGERS [TR] - Sensitive records visible to the wrong users - A new table or department needs restricted access - Audit or privacy requirements for student data	10. YOUR SOLUTION [SL] - Script-controlled ACLs on custom table u_institution_details - Read ACL: role bb1 + data condition Branch is EEE + script; admin allowed by script - Create, Write and Delete ACLs: roles bb2, bb3, bb4 - Tested by impersonating EEE User, Admin and a user with no roles	8.1 ONLINE – CHANNELS OF BEHAVIOUR [CH] - Browsing lists and forms in the ServiceNow web interface - Using the Application Navigator and list filters - Impersonating users to test access

4. EMOTIONS: BEFORE / AFTER [EM] • Before: worried about data exposure, unsure who can see what • After: confident and in control, each user sees only what they should	• Fits the constraints: built-in features and simple scripts only	8.2 OFFLINE – CHANNELS OF BEHAVIOUR [CH] • Team or admin discussions about who gets which role • Access requests raised to administrators • Documented test steps shared with learners
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**Project Design Phase
Proposed Solution**

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Proposed Solution :

Project team shall fill the following information in the proposed solution .

S.No.	Parameter	Description
1.	Problem Statement (Problem to be solved)	In many organisations, anyone who can open a table can see all of its records, including sensitive ones. Standard role-based access is too coarse to restrict records by field value, such as a student's branch. Access to create, edit and delete records is often bundled with read access, so users get more permission than they need.
2.	Idea / Solution description	Implement script-controlled Access Control Lists (ACLs) in ServiceNow on a custom table, u_institution_details, so that access is decided by role, record data and script together. Read ACL: only users with role bb1 see records where Branch is EEE; admins keep full access. Create, Write and Delete ACLs use separate roles (bb2, bb3, bb4), so each permission is granted independently. The solution is verified by impersonating an EEE user, an admin and a user with no roles.
3.	Novelty / Uniqueness	Combines three checks (role, data condition and server-side script) in one ACL instead of relying on roles alone. Separates read, create, write and delete into individual roles, giving least-privilege access at record level.

		Uses only built-in platform features, with no external tools or plugins.
4.	Social Impact / Customer Satisfaction	Protects personal data such as student names, emails and phone numbers from unauthorised viewing or changes. Users see only the records relevant to them, which gives a cleaner, simpler experience. Supports privacy and data-protection expectations for institutions handling student records.
5.	Business Model (Revenue Model)	This is an internal security solution rather than a product sold on its own. Its value comes from reduced risk of data exposure and lower manual access-management effort. It can be offered as a reusable configuration or implementation service for institutions and enterprises on ServiceNow, and as training material for ACL security.
6.	Scalability of the Solution	The same ACL pattern can be applied to other tables and to other fields such as department, region or status. New branches or roles are added by creating roles and ACLs, with no redesign. Runs on the ServiceNow platform, which handles growth in users and records. Script ACLs should stay simple to avoid slowing large lists.

Project Design Phase Solution Architecture

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Solution Architecture:

The business problem is that sensitive records must not be visible to everyone who can open a table. In this project, only users who hold the required role may view records whose Branch field is EEE, while administrators keep full access. Creating, editing and deleting records must also be controlled separately.

The solution is built on the ServiceNow platform using script-controlled Access Control Lists (ACLs). Each ACL combines three checks, all of which must pass before access is granted:

- **Role check** – the user must hold the required custom role (bb1 to bb4, or admin).
- **Data condition** – the record must satisfy the condition (for example, Branch is EEE).
- **ACL script** – a server-side script decides true or false using `gs.hasRole()` and record values.

Solution Architecture Diagram:

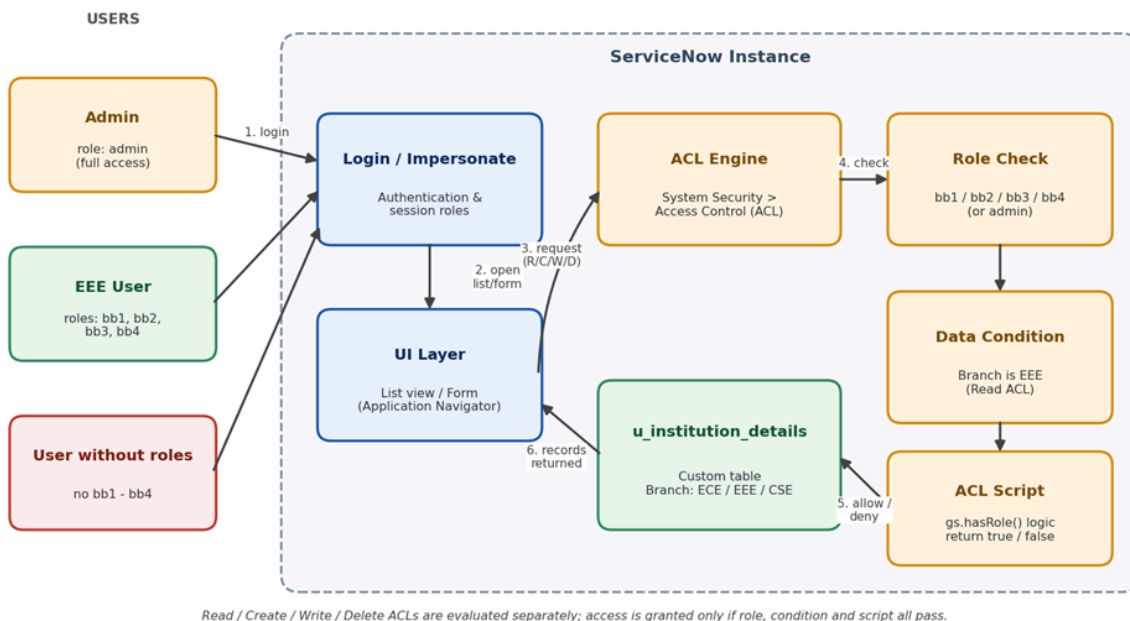


Figure 1: Architecture and data flow of the script-controlled ACL solution

Components:

Component	Purpose	Configuration
Users	EEE User (test user), Admin, and a user with no roles for negative testing	User Administration > Users
Roles	bb1 = read, bb2 = create, bb3 = write, bb4 = delete	User Administration > Roles; all four assigned to EEE User
Custom table	Stores institution/student records	u_institution_details; Branch choice field (ECE, EEE, CSE)
Read ACL	Only bb1 users see EEE records; admin sees all	Advanced ACL with role, data condition and script
Create ACL	Controls the New button	Role bb2, no condition
Write ACL	Controls editing of records	Role bb3, no condition
Delete ACL	Controls deletion of records	Role bb4, no condition
security_admin role	Required to create or edit ACLs	Elevate the role before creating ACLs

Data Flow:

1. The user logs in (or is impersonated) and the session receives the user's roles.
2. The user opens the u_institution_details list or form.
3. The platform sends the request (read, create, write or delete) to the ACL engine.
4. The matching ACL checks the role, then the data condition, then runs the script.
5. If all checks pass, access is allowed; otherwise it is denied.
6. Permitted records are returned to the list or form; restricted records are hidden.

Expected Access Outcome:

User	Roles	Result
Admin	admin	Views all records regardless of branch
EEE User	bb1	Views only EEE branch records
EEE User	bb1 + bb2	Also sees the New button
EEE User	bb1 + bb2 + bb3	Can also edit EEE records

EEE User	bb1 + bb2 + bb3 + bb4	Can also delete EEE records
User without roles	None	Cannot view any records

Technology Stack:

- **Platform:** ServiceNow (Personal Developer Instance)
- **Security:** Access Control (ACL) rules with server-side JavaScript
- **Configuration:** Tables, roles, users, list and form views, impersonation for testing

Reference: ServiceNow Product Documentation – Access Control Rules